EXP 13	Book Return Workflow

Aim:

To record the return of a book, increase the stock by one, mark the record as Returned and prevent the same book from being returned twice.

Theory:

A return is recorded by editing the issue record and entering the Return Date.

- **Validate Book Return (on edit, validate):** Member and Book cannot be changed; an already returned record is locked; Return Date cannot be before Issue Date; on a valid return Available Copies is increased by 1 (never above Total Copies).
- **Update Status on Return (on edit, success):** Status = Returned and Remarks = "Returned on time" or "Returned late (n days after due date)".
- **Restore Stock on Delete (on delete, validate):** If an issue that was never returned is deleted, the copy is given back so the stock stays correct. Deleting a returned record does not change the stock.

Procedure:

1. Create Validate Book Return (Book Issues, Edited, Validations on form submission).
2. Create Update Status on Return (Edited, Successful form submission).
3. Create Restore Stock on Delete (Deleted, Validations on record deletion).
4. In the Issue Register edit issue 3 (Karthik R - 1984), enter Return Date 20-Oct-2026 and click Update.
5. Edit the same record again, enter another return date and click Update.

Program (Validate Book Return):

```
// Book Issues - validation when an issue record is edited (return of a book)
prev = Book_Issues[ID == input.ID];
if(prev.Book != input.Book || prev.Member != input.Member)
{
    alert "Member and Book cannot be changed after the book is issued.";
    cancel submit;
}
if(prev.Issue_Status == "Returned")
{
    alert "This book was already returned on " + prev.Return_Date + ". The return
cannot be recorded again.";
    cancel submit;
}
if(input.Issue_Status == "Returned" && input.Return_Date == null)
{
    alert "Enter the Return Date to mark the book as Returned.";
    cancel submit;
}
if(input.Return_Date != null && input.Return_Date < input.Issue_Date)
{
    alert "Return Date cannot be before the Issue Date.";
    cancel submit;
}
// valid return: increase stock by exactly one (once only - a returned record is locked
```

```
above)
if(input.Return_Date != null)
{
    bk = Books[ID == input.Book];
    if(bk.Available_Copies < bk.Total_Copies)
    {
        bk.Available_Copies = bk.Available_Copies + 1;
    }
}
```

Program (Update Status on Return / Restore Stock on Delete):

```
// Book Issues - after a successful edit: set the correct status
if(input.Return_Date != null)
{
    input.Issue_Status = "Returned";
    if(input.Return_Date > input.Due_Date)
    {
        input.Single_Line = "Returned late (" +
input.Due_Date.daysBetween(input.Return_Date) + " days after due date)";
    }
    else
    {
        input.Single_Line = "Returned on time";
    }
}
else if(input.Due_Date < zoho.currentdate)
{
    input.Issue_Status = "Overdue";
}
else
{
    input.Issue_Status = "Issued";
}

// Book Issues - before deletion: give the copy back if the book was never returned
if(input.Issue_Status != "Returned")
{
    bk = Books[ID == input.Book];
    if(bk.Available_Copies < bk.Total_Copies)
    {
        bk.Available_Copies = bk.Available_Copies + 1;
    }
}
```

Output:

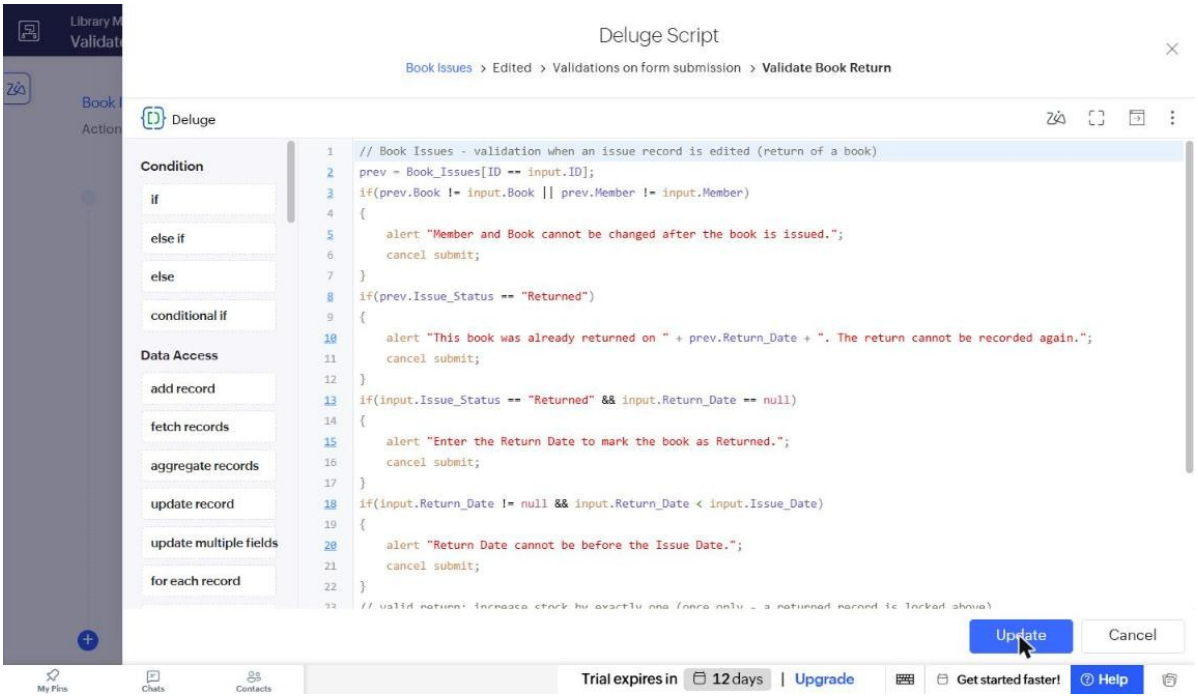


Fig. 13.1: Validate Book Return script

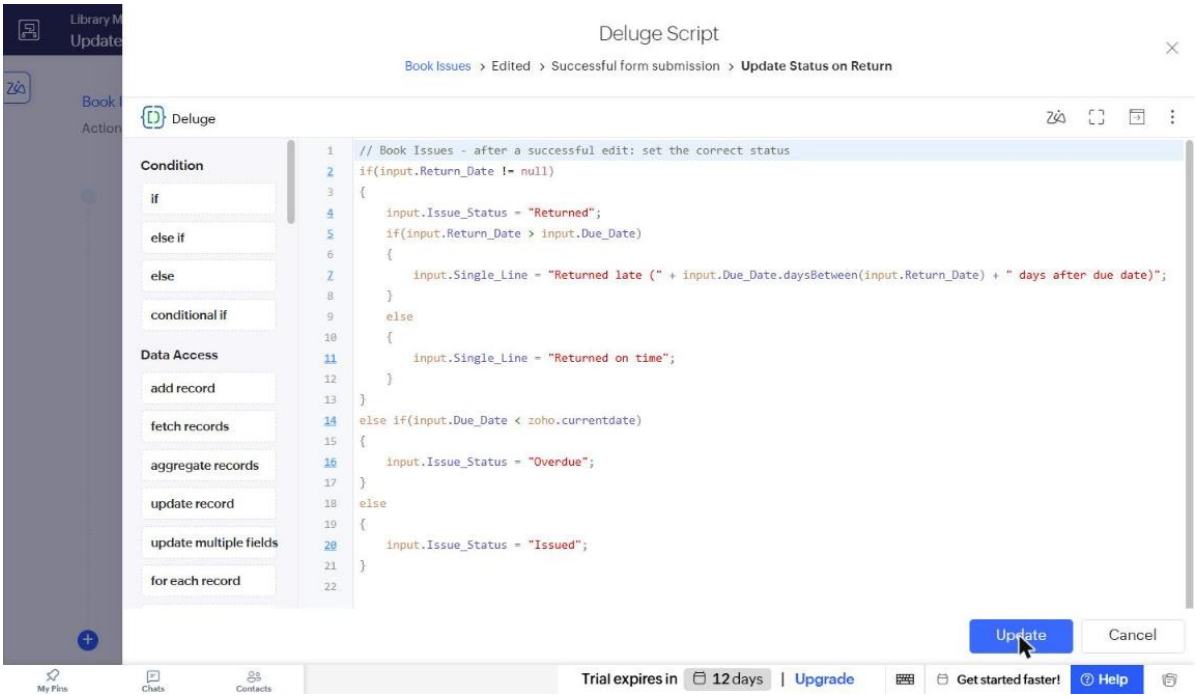


Fig. 13.2: Update Status on Return script

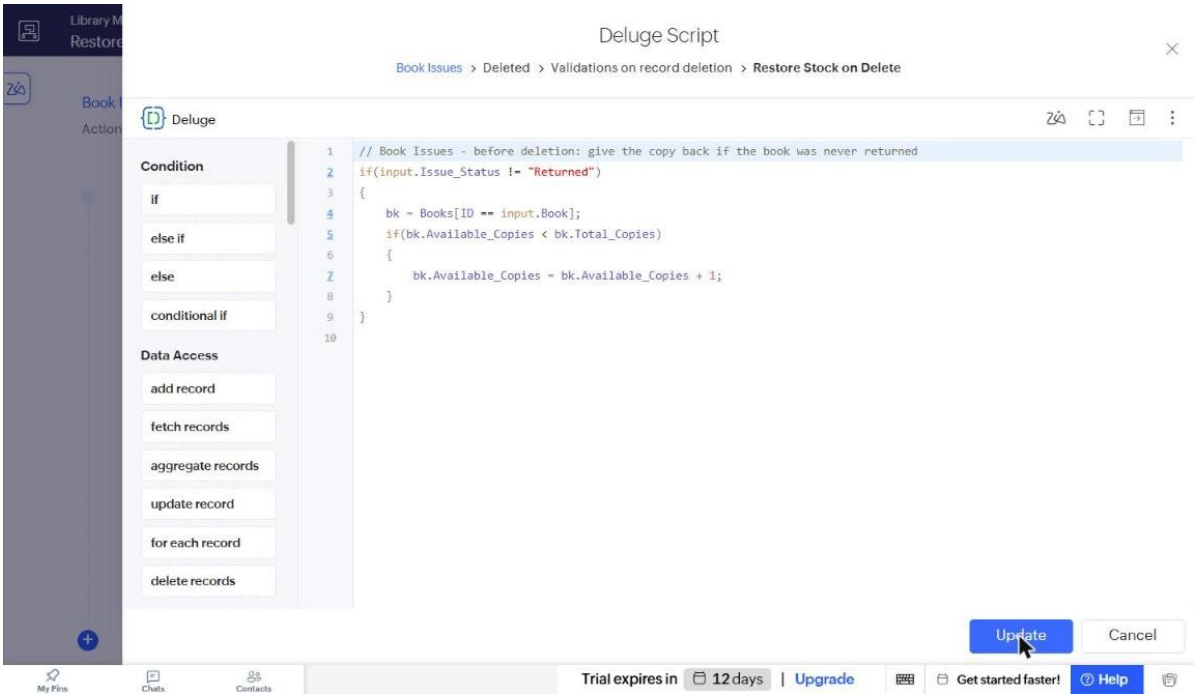


Fig. 13.3: Restore Stock on Delete script

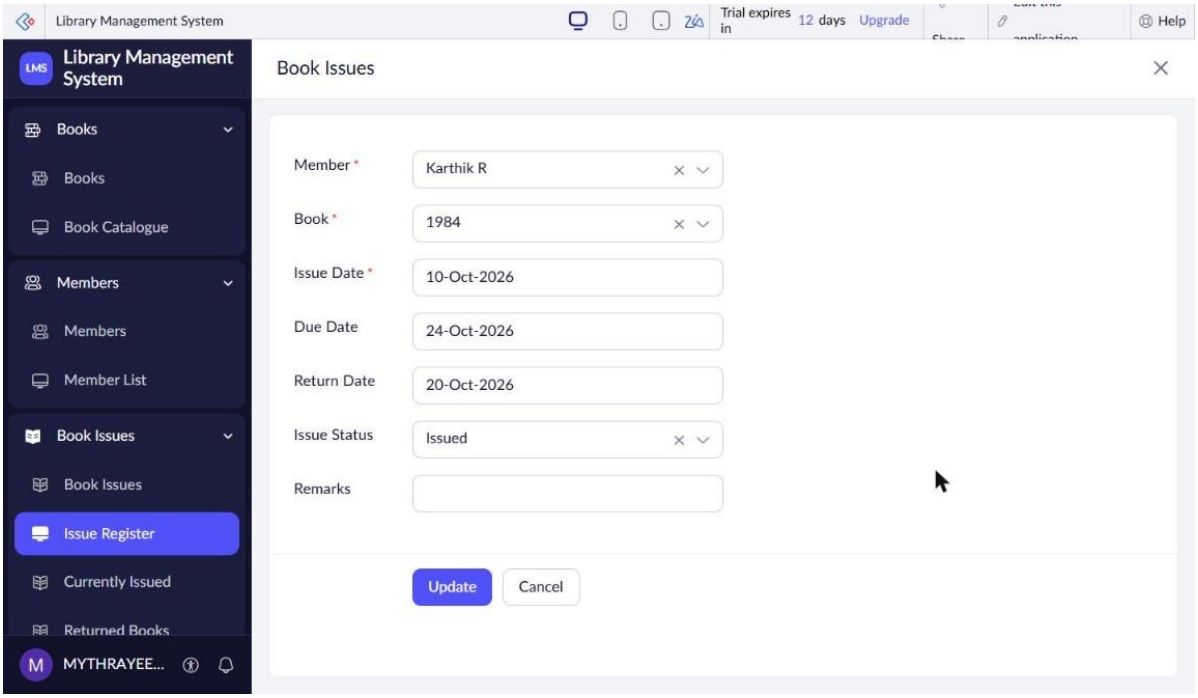


Fig. 13.4: Return Date entered for 1984

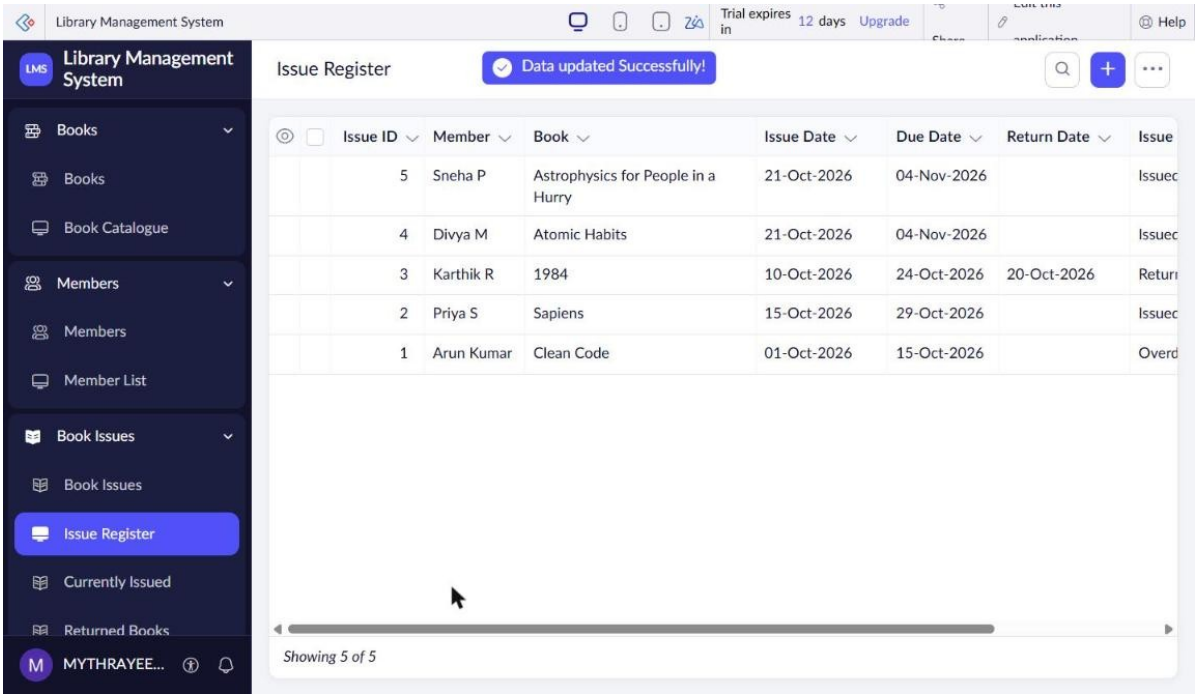


Fig. 13.5: Record updated

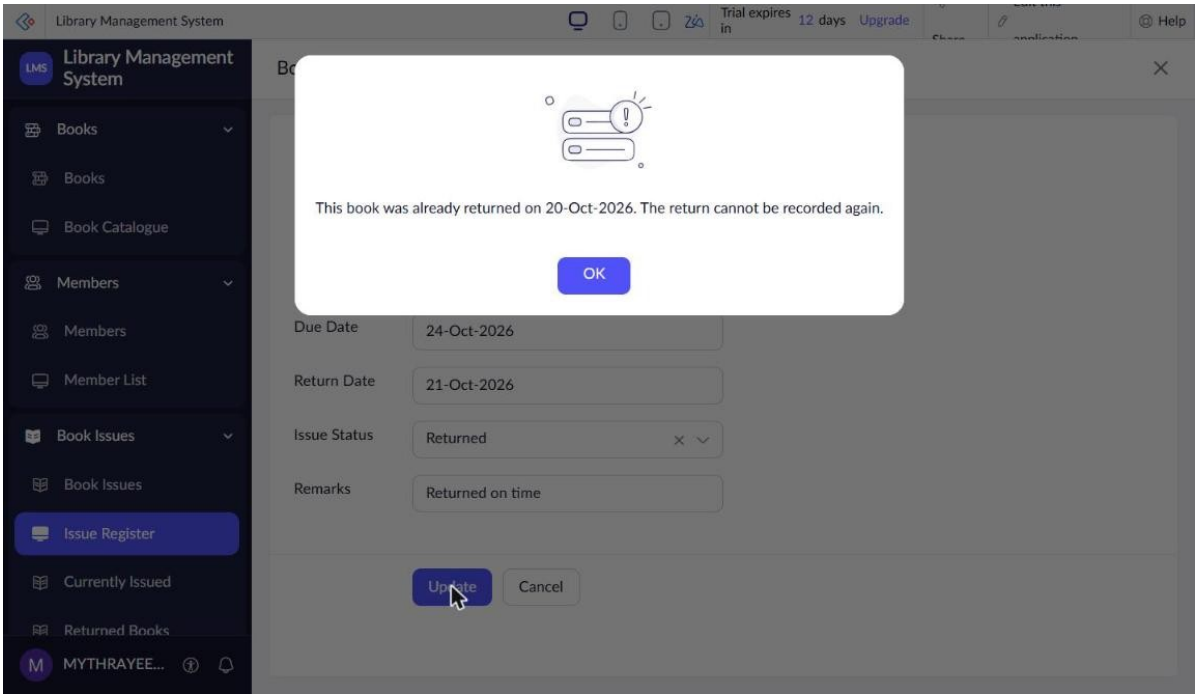


Fig. 13.6: Second return of the same book blocked

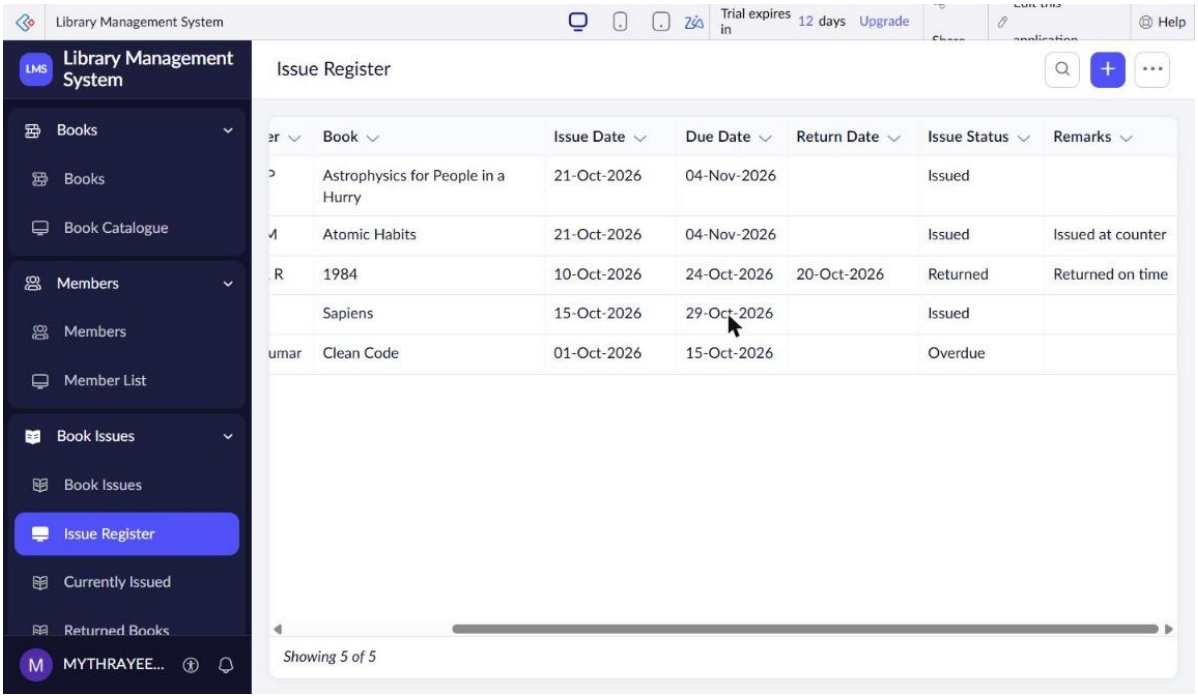


Fig. 13.7: Status and remarks after return

Observation:

Action	Result	1984 Available Copies
Before return	Status Issued	0
Return on 20-Oct-2026	Status Returned, Remarks "Returned on time"	1
Return again on 21-Oct-2026	Blocked - already returned on 20-Oct-2026	1 (unchanged)

Result:

The return workflow increases the stock by exactly one, marks the issue as Returned with a remark, and blocks a second return of the same record.